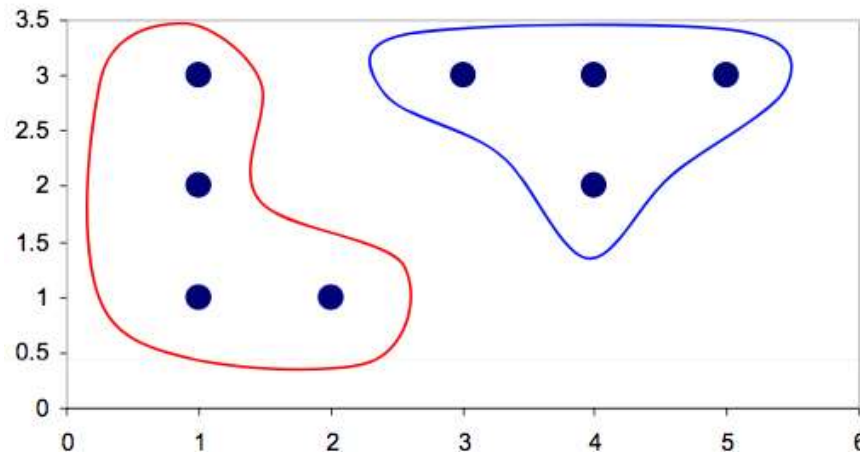


Algorithms: Clustering

Tilani Gunawardena

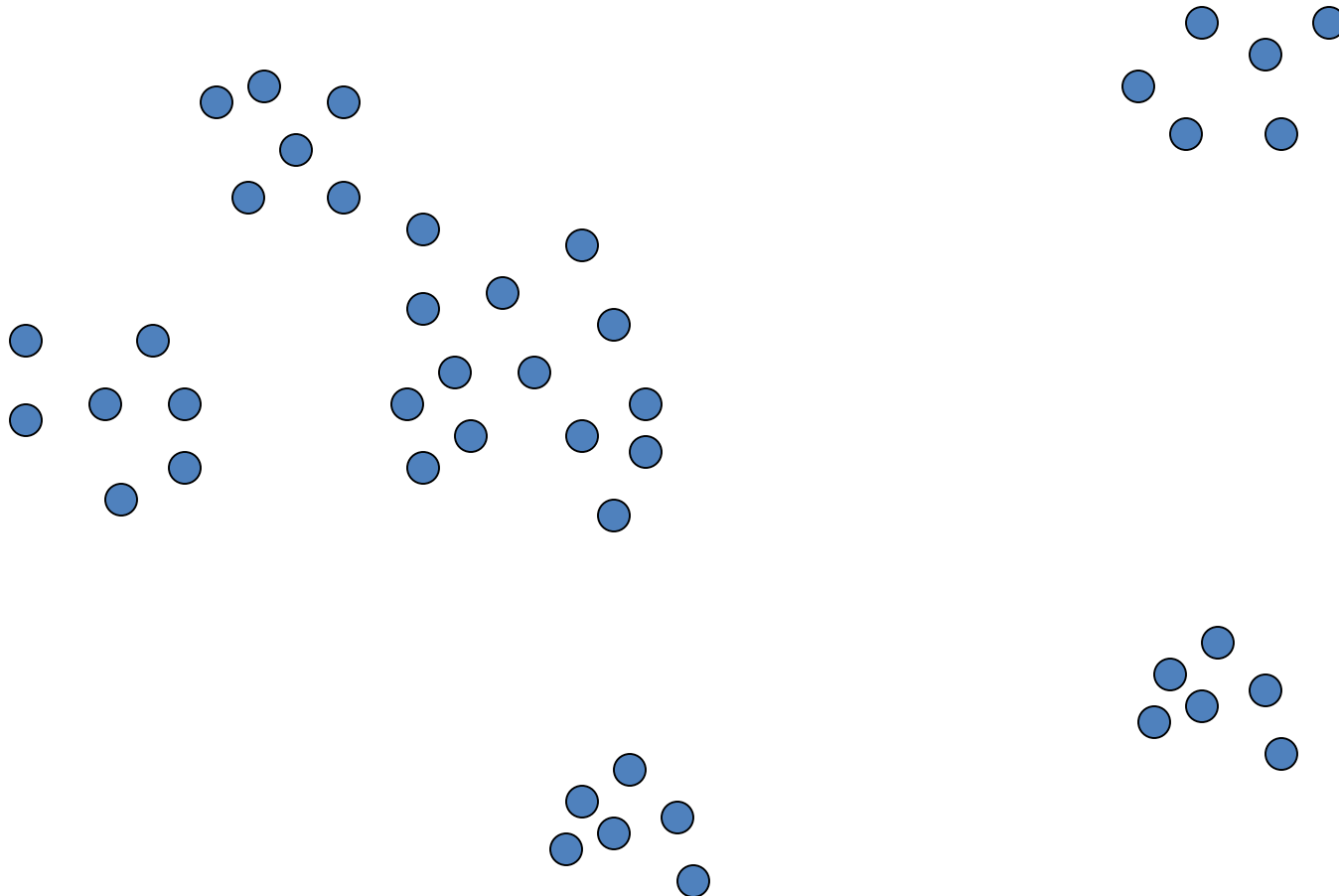
What is Clustering?

- Grouping of records ,observations or cases into classes of similar objects.

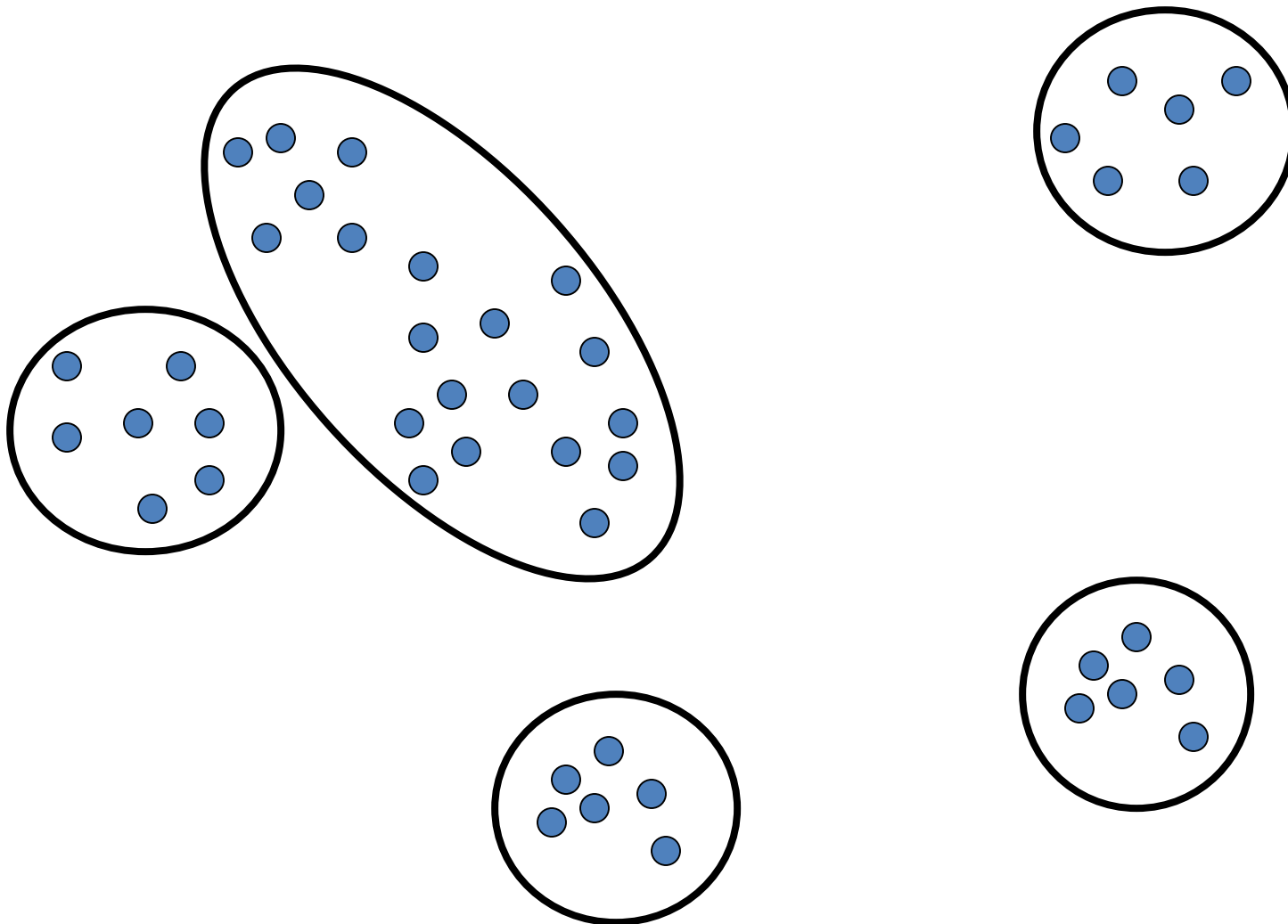


- A cluster is a collection of records,
 - Similar to one another
 - Dissimilar to records in other clusters

Clustering



Clustering



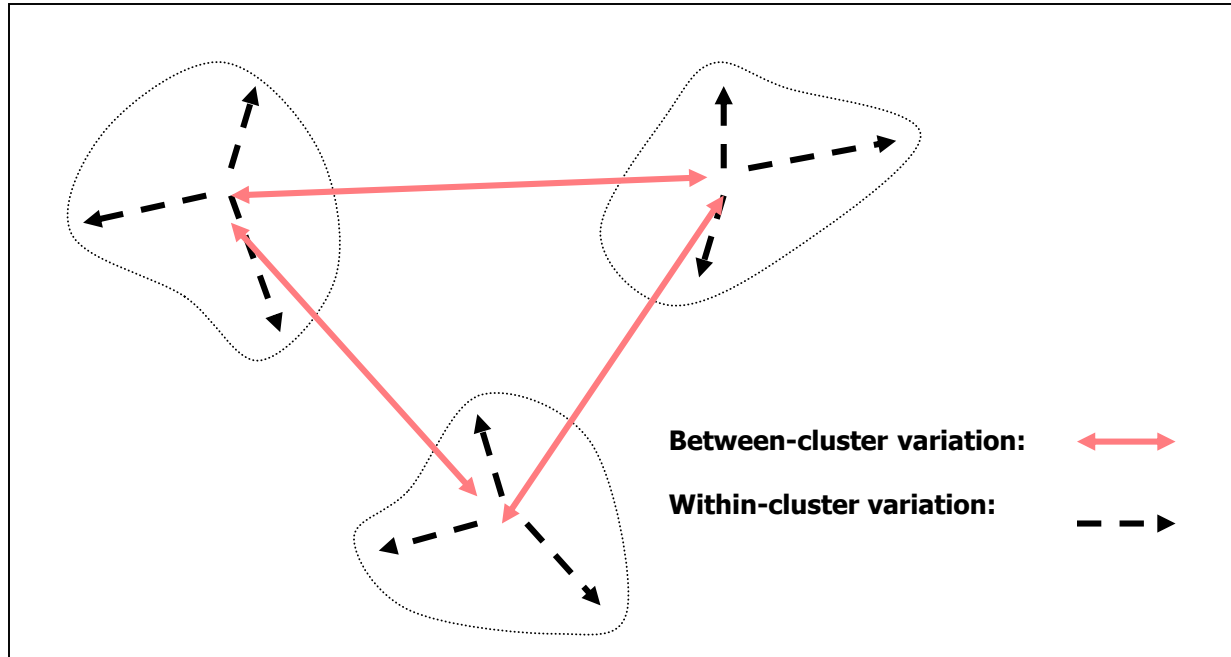
Difference between Clustering and Classification

- There is **no target variable** for clustering
- Clustering does not try to classify or predict the values of a target variable.
- Instead, clustering algorithms seek to segment the entire data set into relatively homogeneous subgroups or clusters,
 - Where the similarity of the records **within the cluster** is **maximized**, and
 - Similarity to records **outside this cluster** is **minimized**.

Goal of Clustering

- Identification of groups of records such that similarity within a group is very high while the similarity to records in other groups is very low.
 - group data points that are close (or **similar**) to each other
 - identify such groupings (or clusters) in an **unsupervised** manner
 - Unsupervised: no information is provided to the algorithm on which data points belong to which clusters
- In other words,
 - Clustering algorithm seeks to construct clusters of records such that the **between-cluster variation**(BCV) is large compared to the **within-cluster variation**(WCV)

Goal of Clustering



(Intra-cluster distance) the sum of distances between objects in the same cluster are **minimized**

(Inter-cluster distance) while the distances between different clusters are **maximized**

between-cluster variation(BCV) is large compared to the **within-cluster variation**(WCV)

Clustering

- Clustering techniques apply when there is no class to be predicted
- As we've seen clusters can be:
 - disjoint vs. overlapping
 - deterministic vs. probabilistic
 - flat vs. hierarchical
- *k-means Algorithm*
 - *k-means* clusters are disjoint, deterministic, and flat

Issues Related to Clustering

- How to measure similarity
 - Euclidian Distance
 - City-block Distance
 - Minkowski Distance
- How to recode categorical variables?
- How to standardize or normalize numerical variables?
 - Min-Max Normalization
 - Z-score standardization (ms)
- How many clusters we expect to uncover?

Type of Clustering

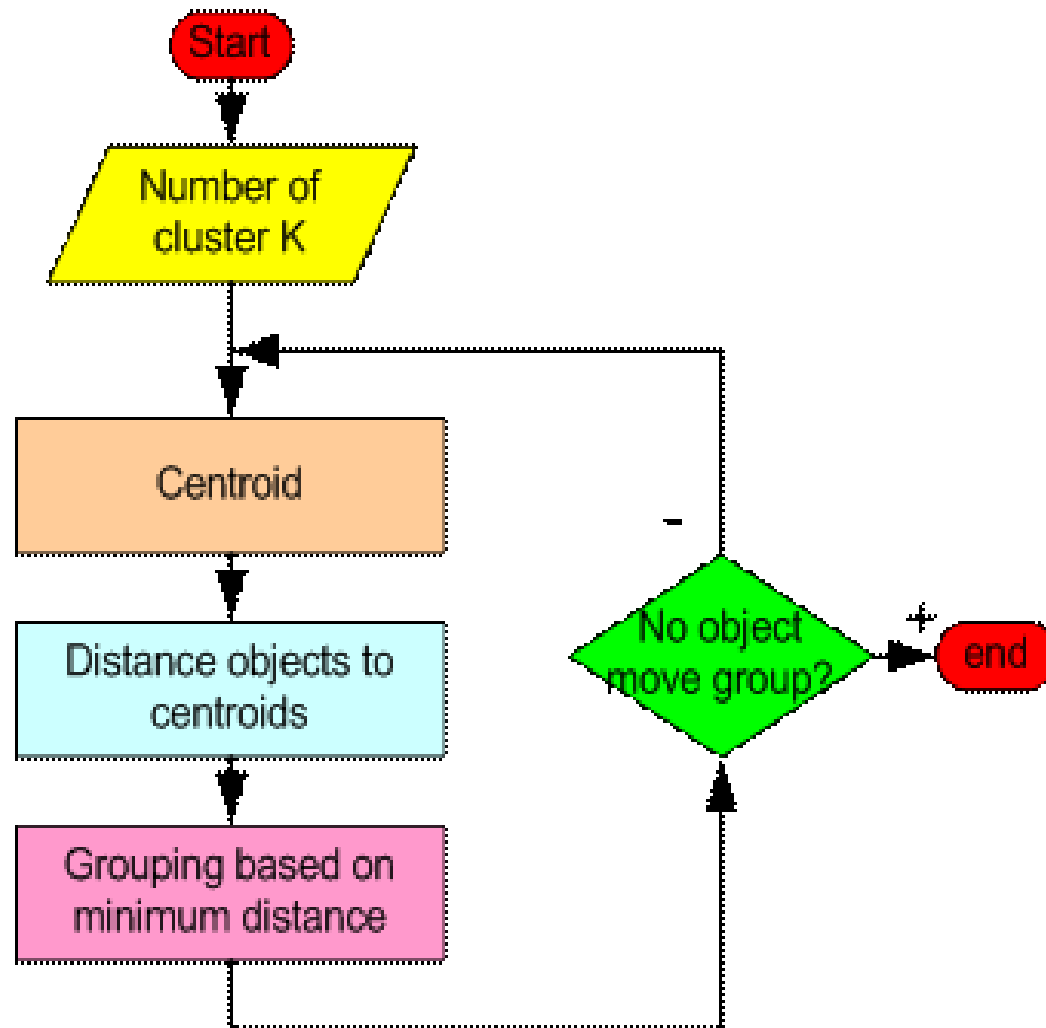
- **Partitional clustering:** Partitional algorithms determine all clusters at once. They include:
 - K-Means Clustering
 - Fuzzy c-means clustering
 - QT clustering
- **Hierarchical Clustering:**
 - Agglomerative ("**bottom-up**"): Agglomerative algorithms begin with each element as a separate cluster and merge them into successively larger clusters.
 - Divisive ("**top-down**"): Divisive algorithms begin with the whole set and proceed to divide it into successively smaller clusters.

K-means Clustering

k-Means Clustering

- Input: n objects (or points) and a number k
- Algorithm
 - 1) Randomly assign K records to be the initial cluster center locations
 - 2) Assign each object to the group that has the closest centroid
 - 3) When all objects have been assigned, recalculate the positions of the K centroids
 - 4) Repeat steps 2 to 3 until convergence or termination

K-Mean Clustering



Termination Conditions

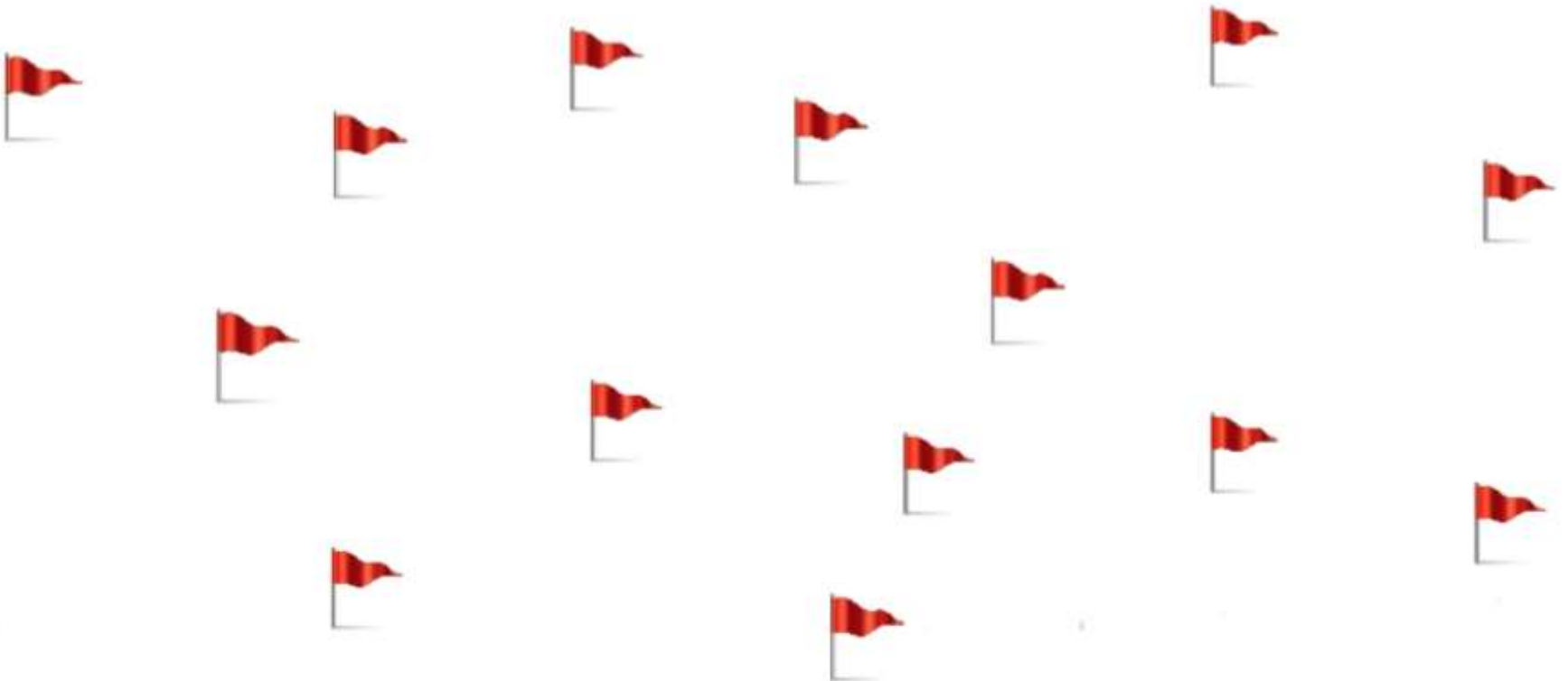
- The algorithm terminates when the **centroids no longer change**.
- The SSE(**sum of squared errors**) value is less than some small **threshold** value α

$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

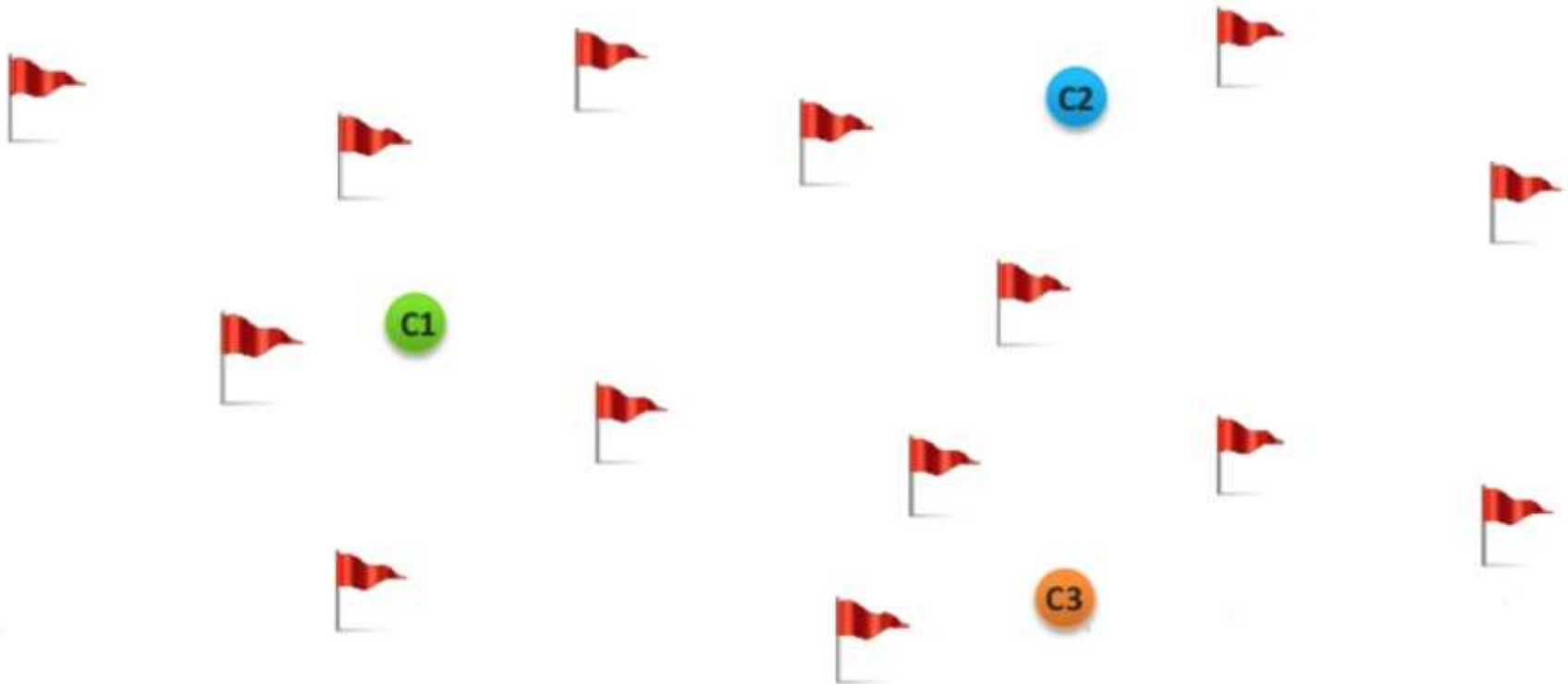
- Where $p \in C_i$ represents each data point in cluster i and m_i represent the centroid of cluster i .

Example 1:

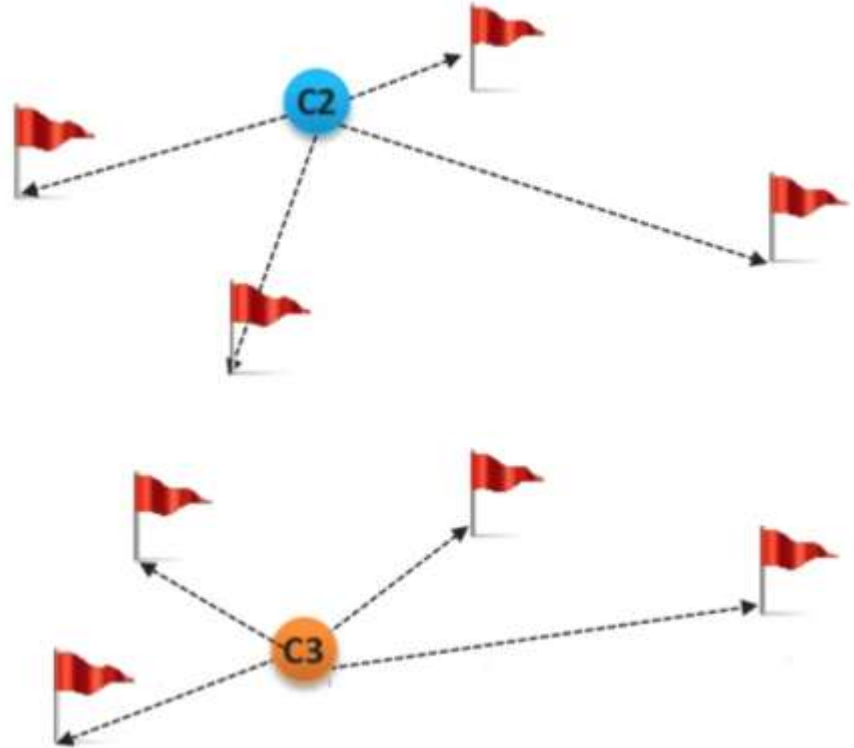
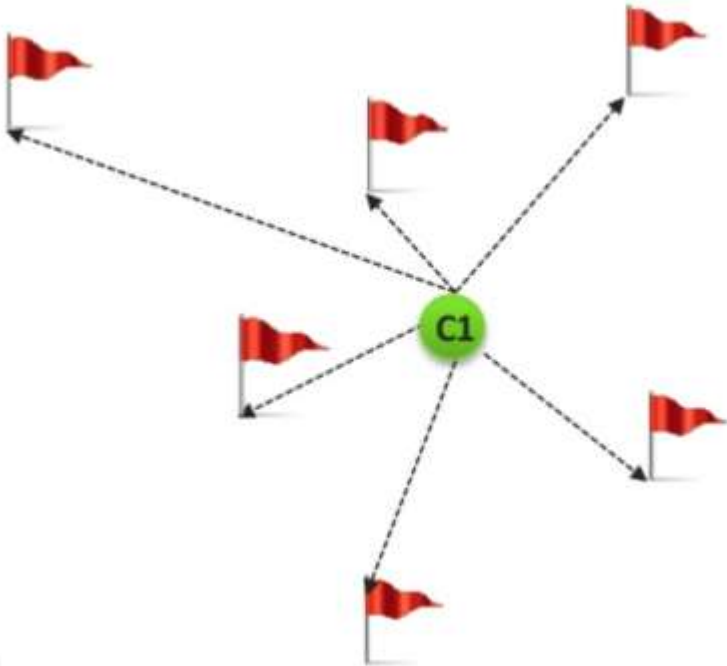
- Lets s suppose the following points are the delivery locations for pizza



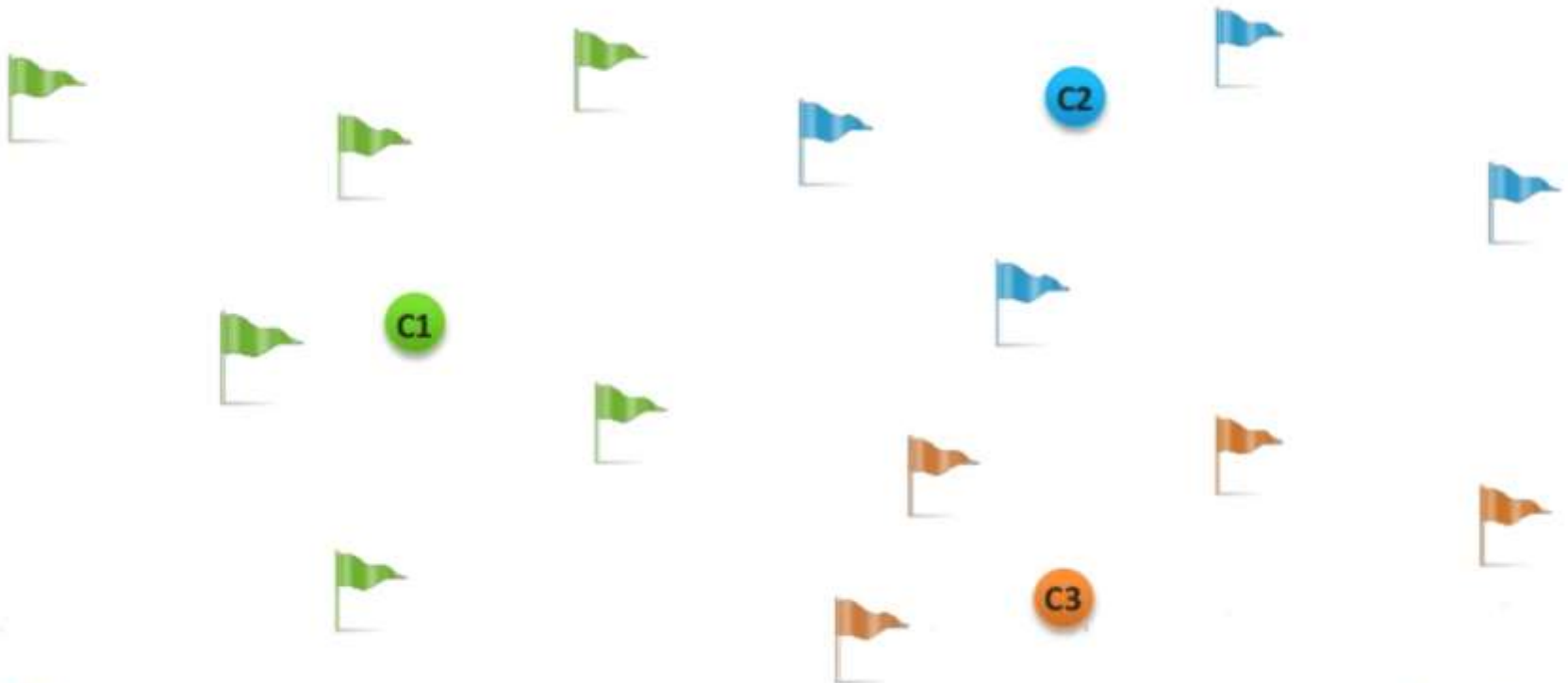
- Lets locate three cluster centers randomly



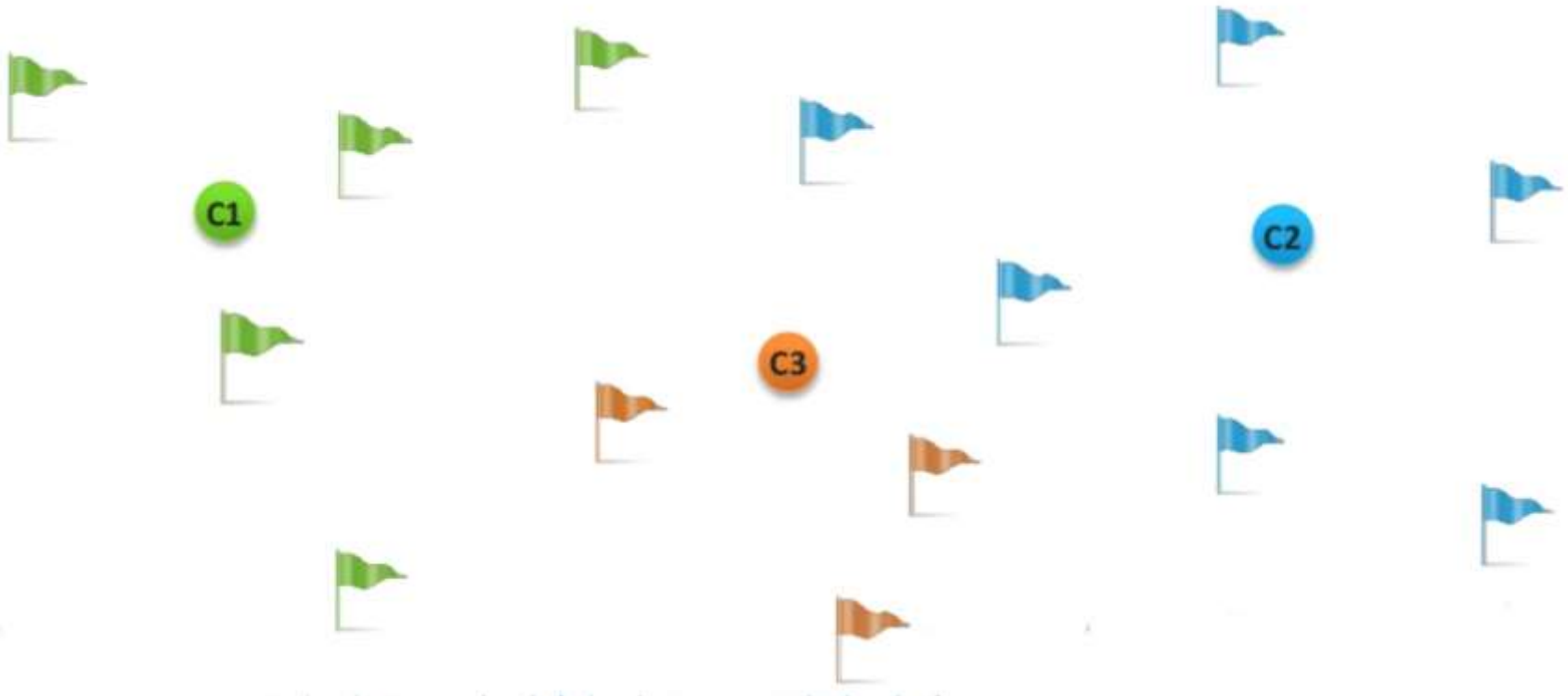
- Find the distance of points as shown



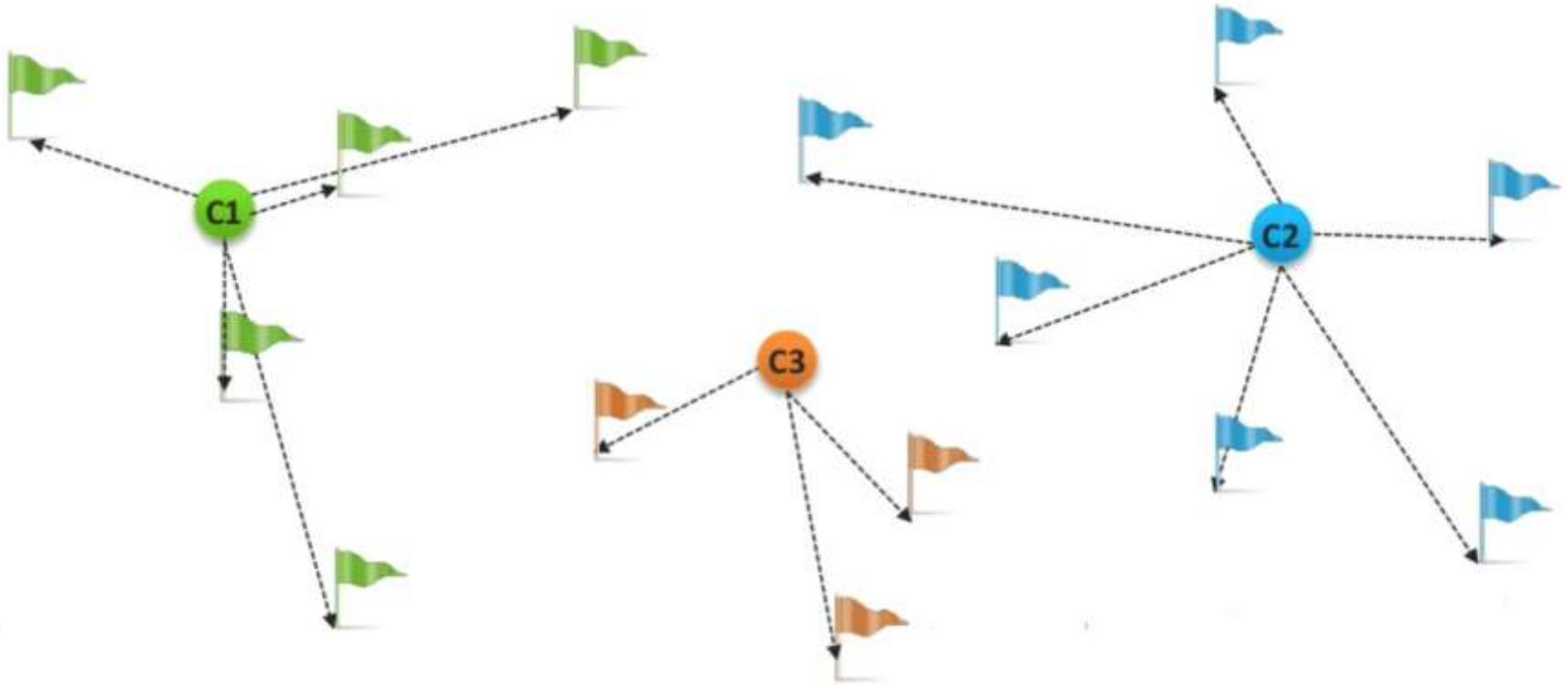
- Assign the points to the nearest cluster center based on the distance between each center and the point



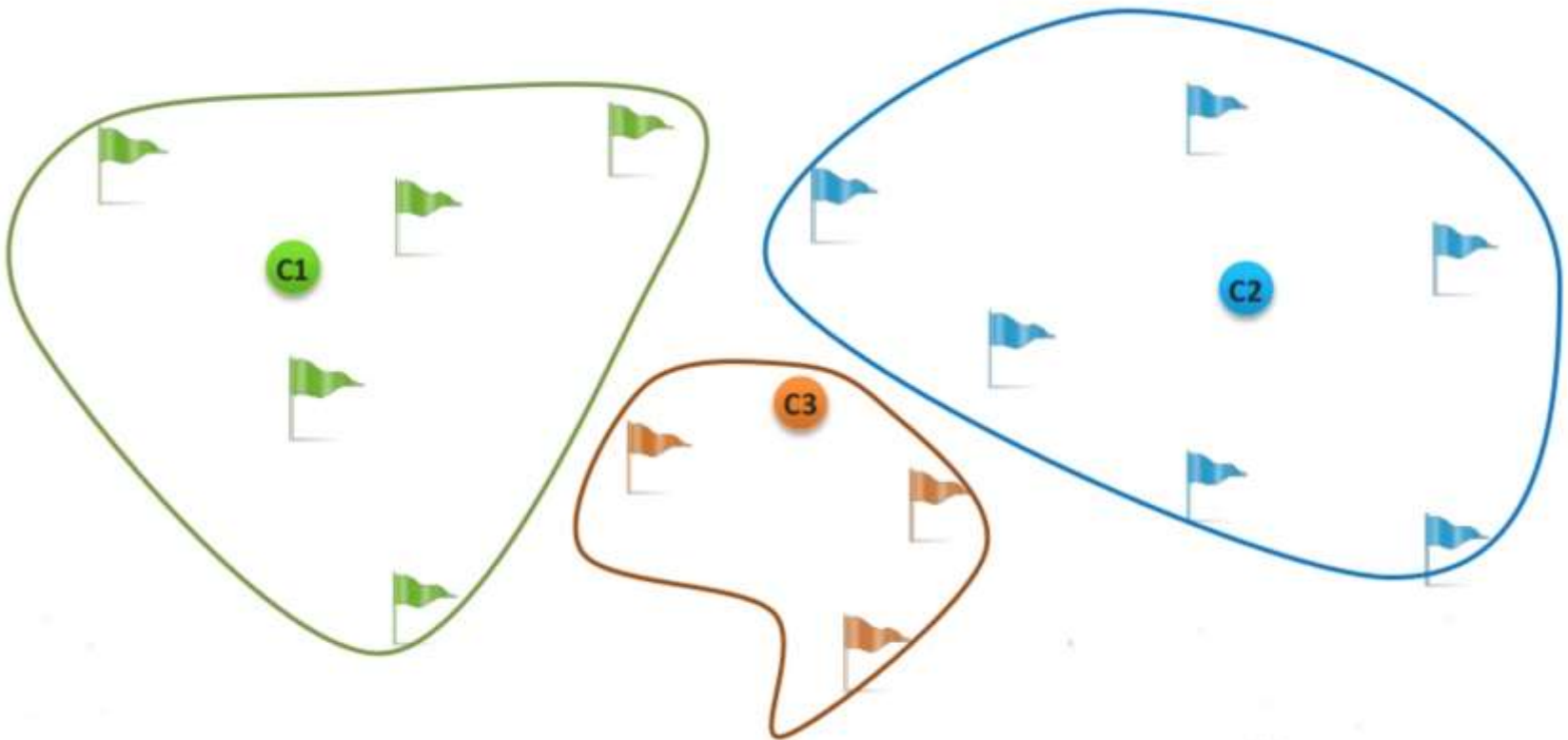
- Re-assign the cluster centres and locate nearest points



- Re-assign the cluster centres and locate nearest points, calculate the distance



Form the three clusters



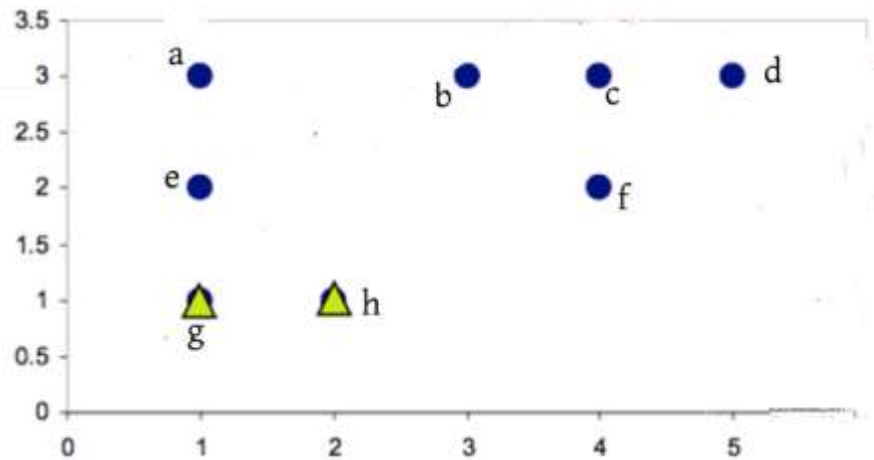
Example 2:

- Suppose that we have eight data points in two-dimensional space as follows

a	b	c	d	e	f	g	h
(1,3)	(3,3)	(4,3)	(5,3)	(1,2)	(4,2)	(1,1)	(2,1)

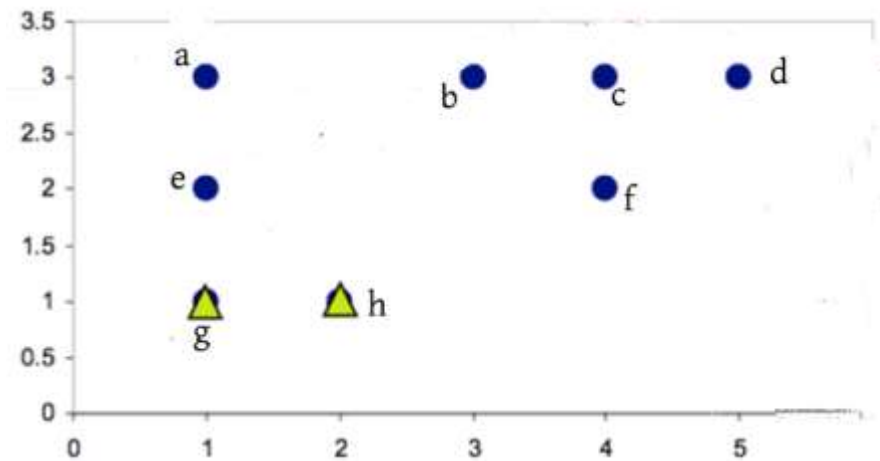
- And suppose that we are interested in uncovering $k=2$ clusters.

$$D(X,Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



Point	Distance from m1	Distance from m2	Cluster membership
a			
b			
c			
d			
e			
f			
g			
h			

$$D(X,Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



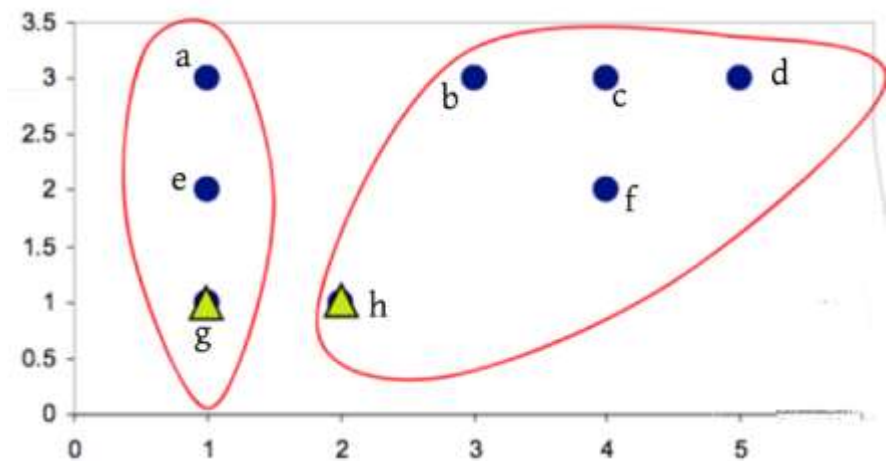
Point	Distance from m1	Distance from m2	Cluster membership
a	2.00	2.24	
b	2.83	2.24	
c	3.61	2.83	
d	4.47	3.61	
e	1.00	1.41	
f	3.16	2.24	
g	0.00	1.00	
h	1.00	0.00	

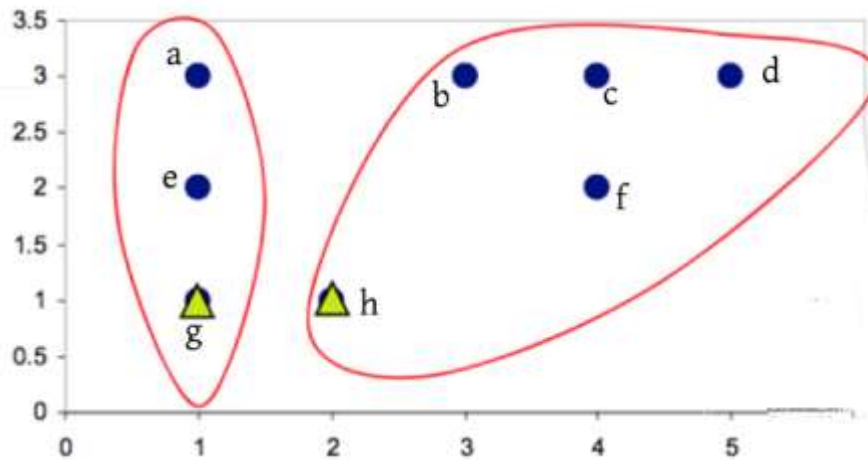
Point	Distance from m1	Distance from m2	Cluster membership
a	2.00	2.24	C1
b	2.83	2.24	C2
c	3.61	2.83	C2
d	4.47	3.61	C2
e	1.00	1.41	C1
f	3.16	2.24	C2
g	0.00	1.00	C1
h	1.00	0.00	C2

$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

$$SSE = 2^2 + 2.24^2 + 2.83^2 + 3.61^2 + 1^2 + 2.24^2 + 0^2 + 0^2 = 36$$

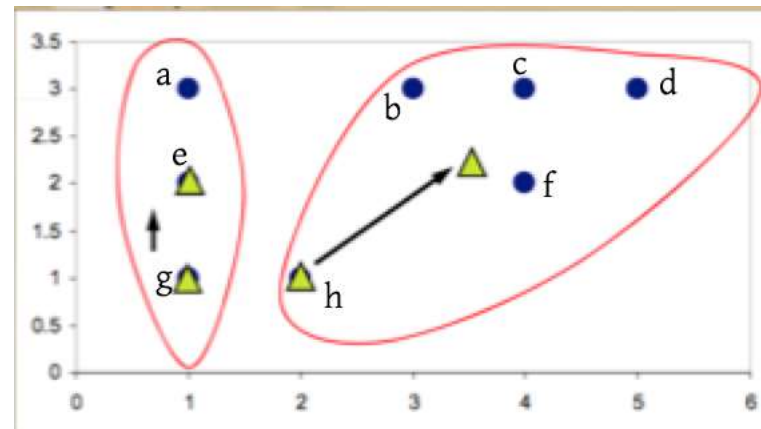
$$d(m1, m2) = 1$$



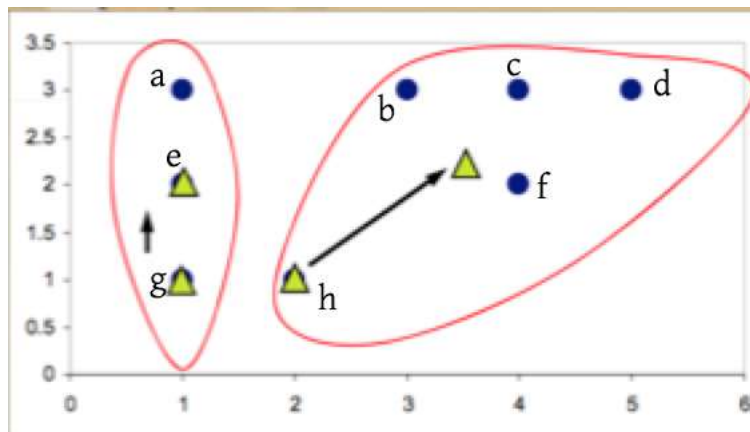


Centroid of the cluster 1 is
 $[(1+1+1)/3, (3+2+1)/3]$
 $= (1, 2)$

Centroid of the cluster 2 is
 $[(3+4+5+4+2)/5, (3+3+3+2+1)/5]$
 $= (3.6, 2.4)$



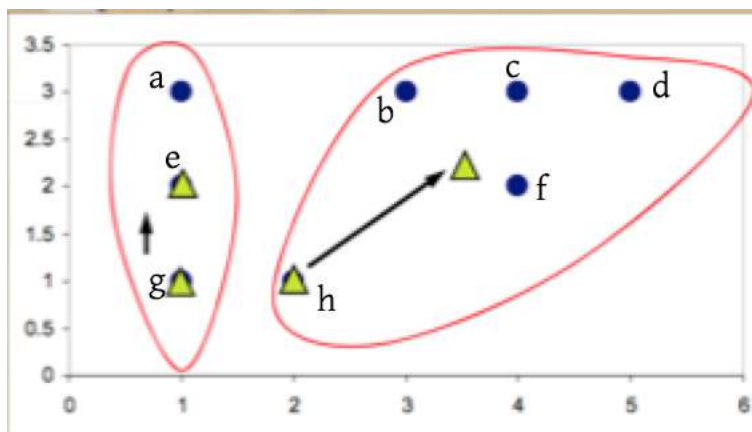
Point	Distance from m1	Distance from m2	Cluster membership
a			
b			
c			
d			
e			
f			
g			
h			



m1=(1,2)

m2=(3.6,2.4)

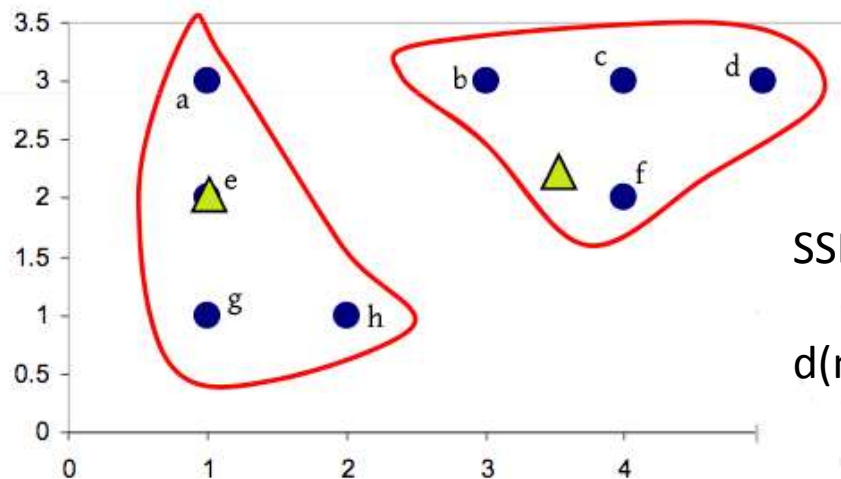
Point	Distance from m1	Distance from m2	Cluster membership
a	1.00	2.67	
b	2.24	0.85	
c	3.61	0.72	
d	4.12	1.52	
e	0.00	2.63	
f	3.00	0.57	
g	1.00	2.95	
h	1.41	2.13	



$m1=(1,2)$

$m2=(3.6,2.4)$

Point	Distance from m1	Distance from m2	Cluster membership
a	1.00	2.67	C1
b	2.24	0.85	C2
c	3.61	0.72	C2
d	4.12	1.52	C2
e	0.00	2.63	C1
f	3.00	0.57	C2
g	1.00	2.95	C1
h	1.41	2.13	C1

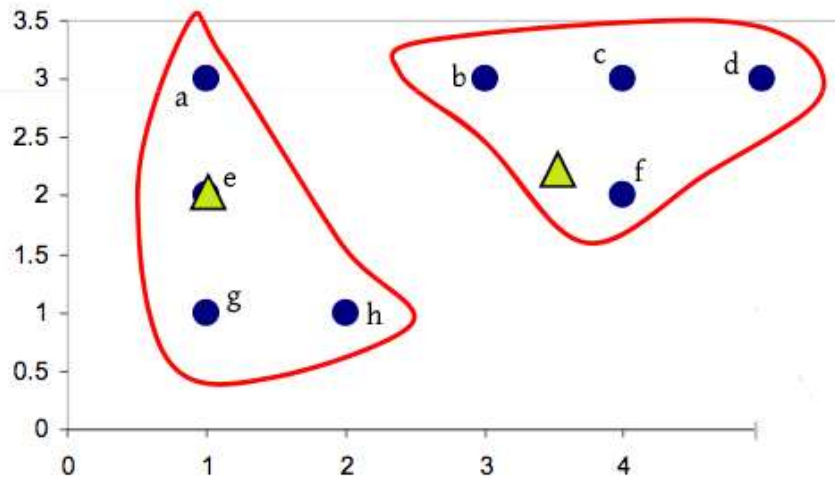


$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

$$SSE = 1^2 + 0.85^2 + 0.72^2 + 1.52^2 + 0^2 + 0.57^2 + 1^2 + 1.41^2 = 7.88$$

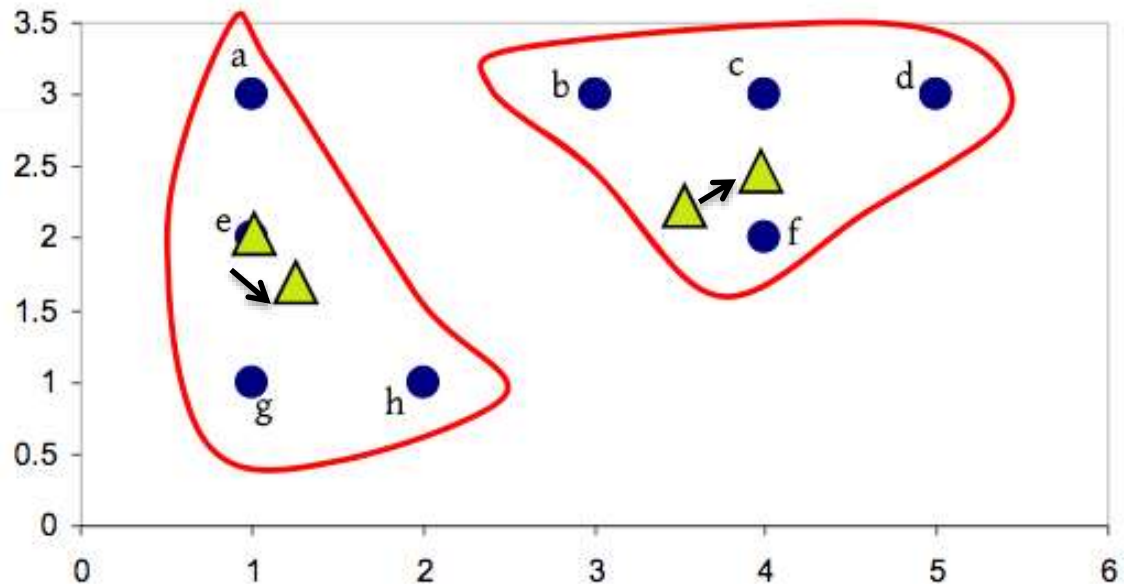
$$d(m1, m2) = 2.63$$

m1(1,2)
m2(3.6,2.4)

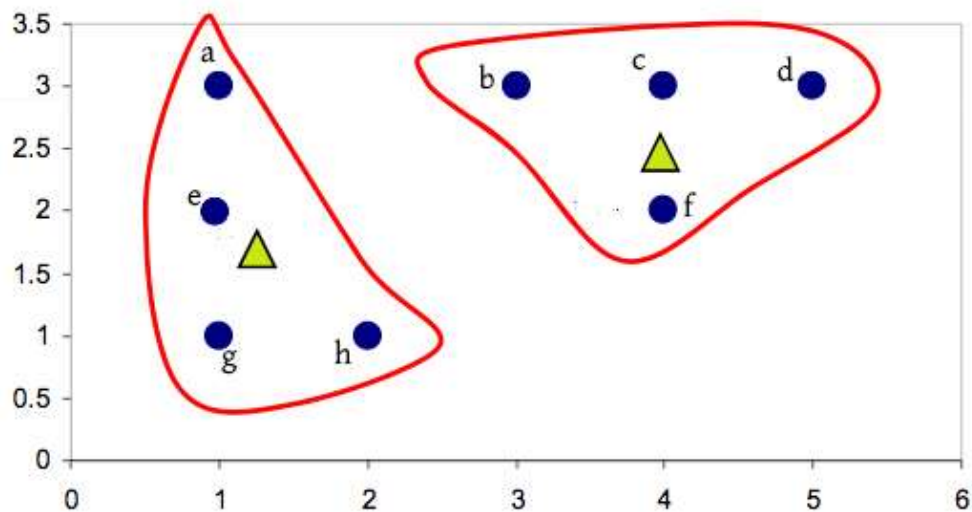


Centroid of the cluster 1 is
 $[(1+1+1+2)/4, (3+2+1+1)/4]$
 $= (1.25, 1.75)$

Centroid of the cluster 2 is
 $[(3+4+5+4)/4, (3+3+3+2)/4]$
 $= (4, 2.75)$



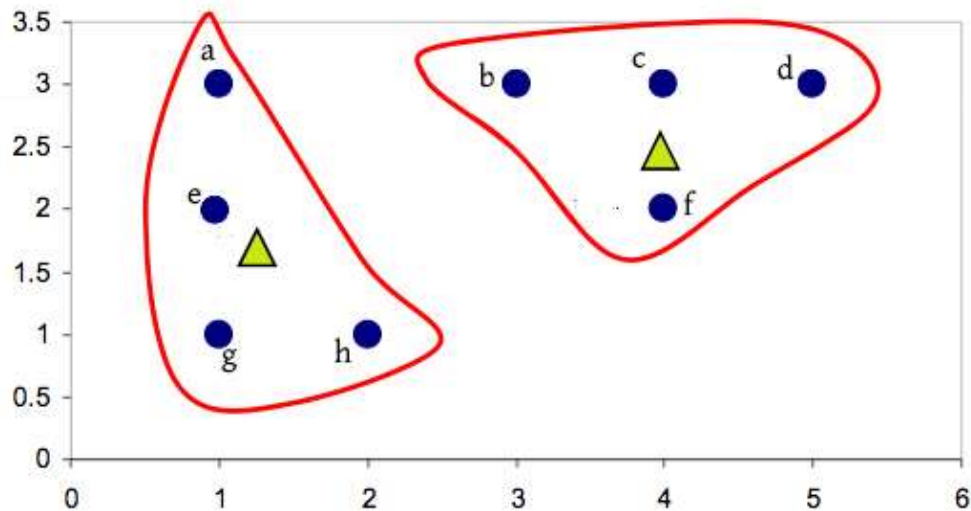
Point	Distance from m1	Distance from m2	Cluster membership
a			
b			
c			
d			
e			
f			
g			
h			



m1(1.25,1.75)

m2(4,2.75)

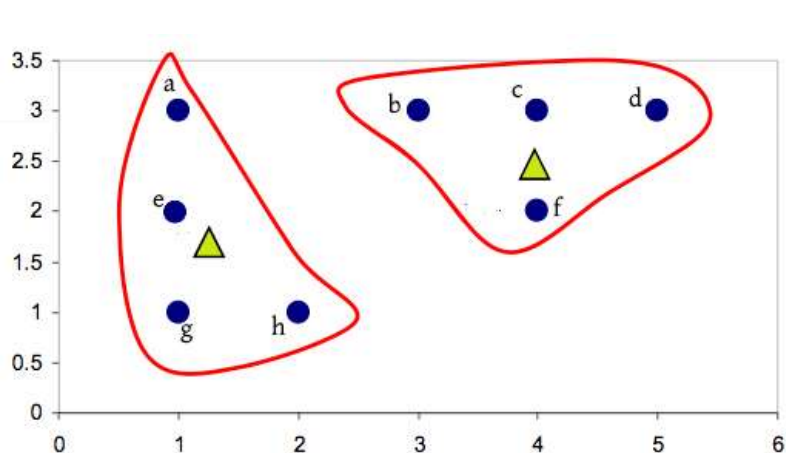
Point	Distance from m1	Distance from m2	Cluster membership
a	1.27	3.01	
b	2.15	1.03	
c	3.02	0.25	
d	3.95	1.03	
e	0.35	3.09	
f	2.76	0.75	
g	0.79	3.47	
h	1.06	2.66	



m1(1.25,1.75)

m2(4,2.75)

Point	Distance from m1	Distance from m2	Cluster membership
a	1.27	3.01	C1
b	2.15	1.03	C2
c	3.02	0.25	C2
d	3.95	1.03	C2
e	0.35	3.09	C1
f	2.76	0.75	C2
g	0.79	3.47	C1
h	1.06	2.66	C1



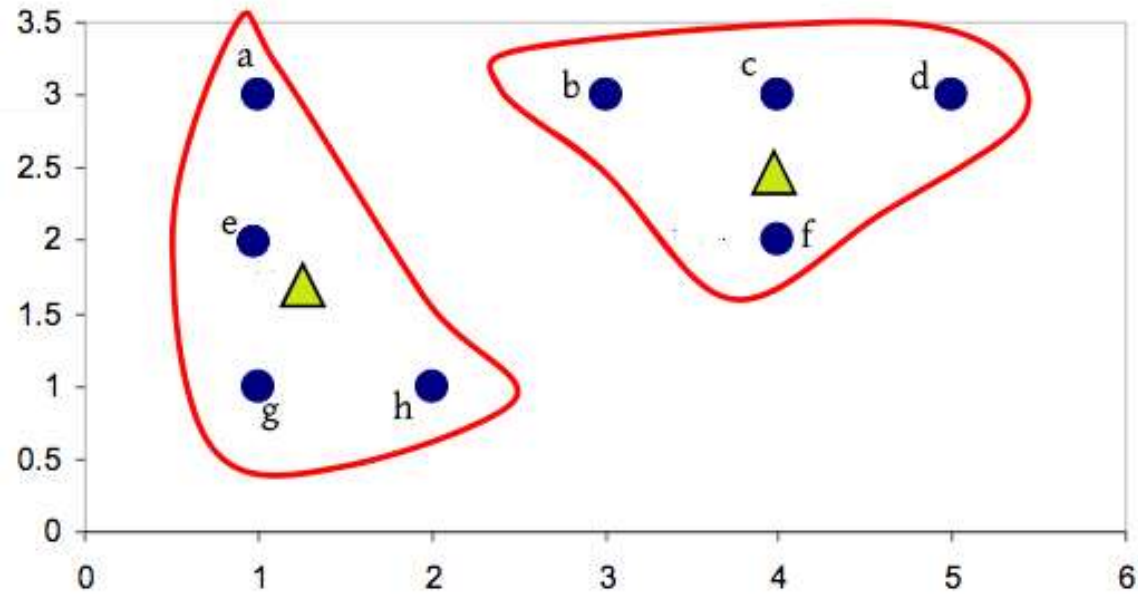
m1(1.25,1.75)

m2(4,2.75)

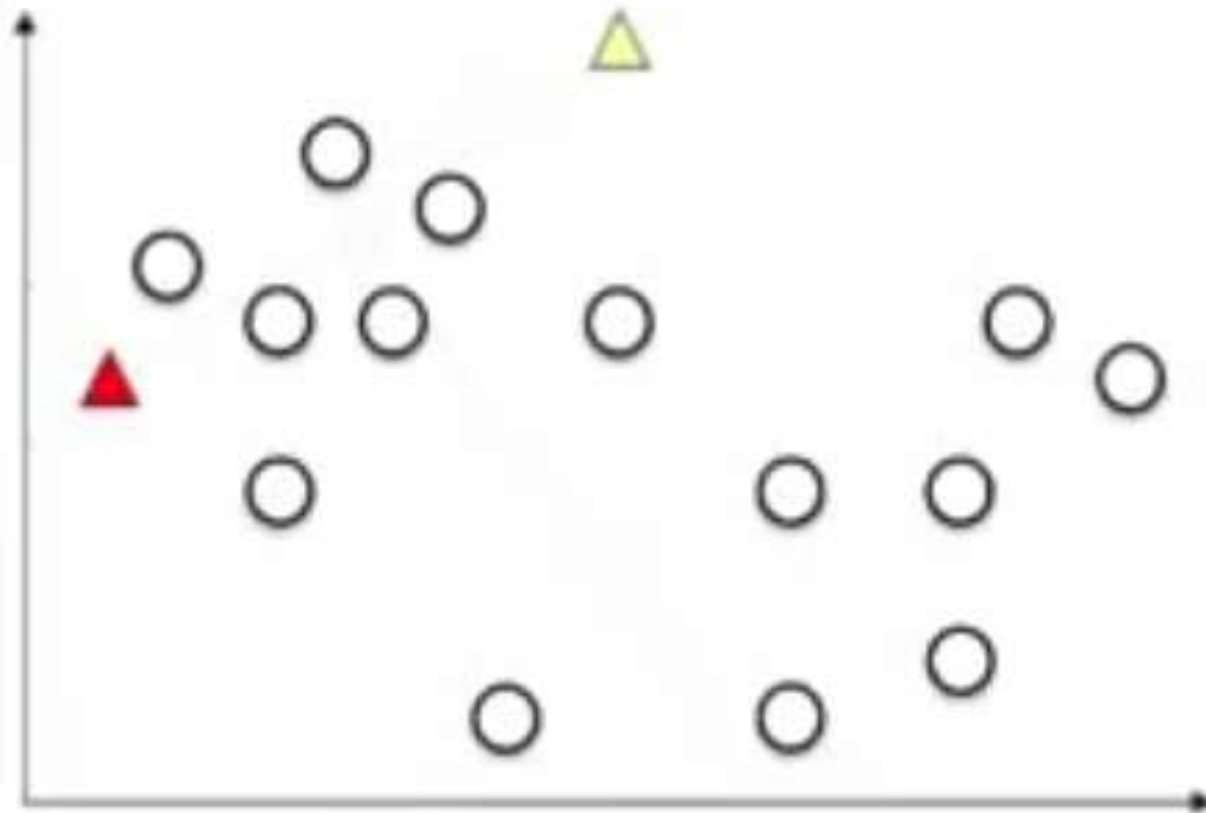
$$SSE = 1.27^2 + 1.03^2 + 0.25^2 + 1.03^2 + 0.35^2 + 0.75^2 + 0.79^2 + 1.06^2 = 6.25$$

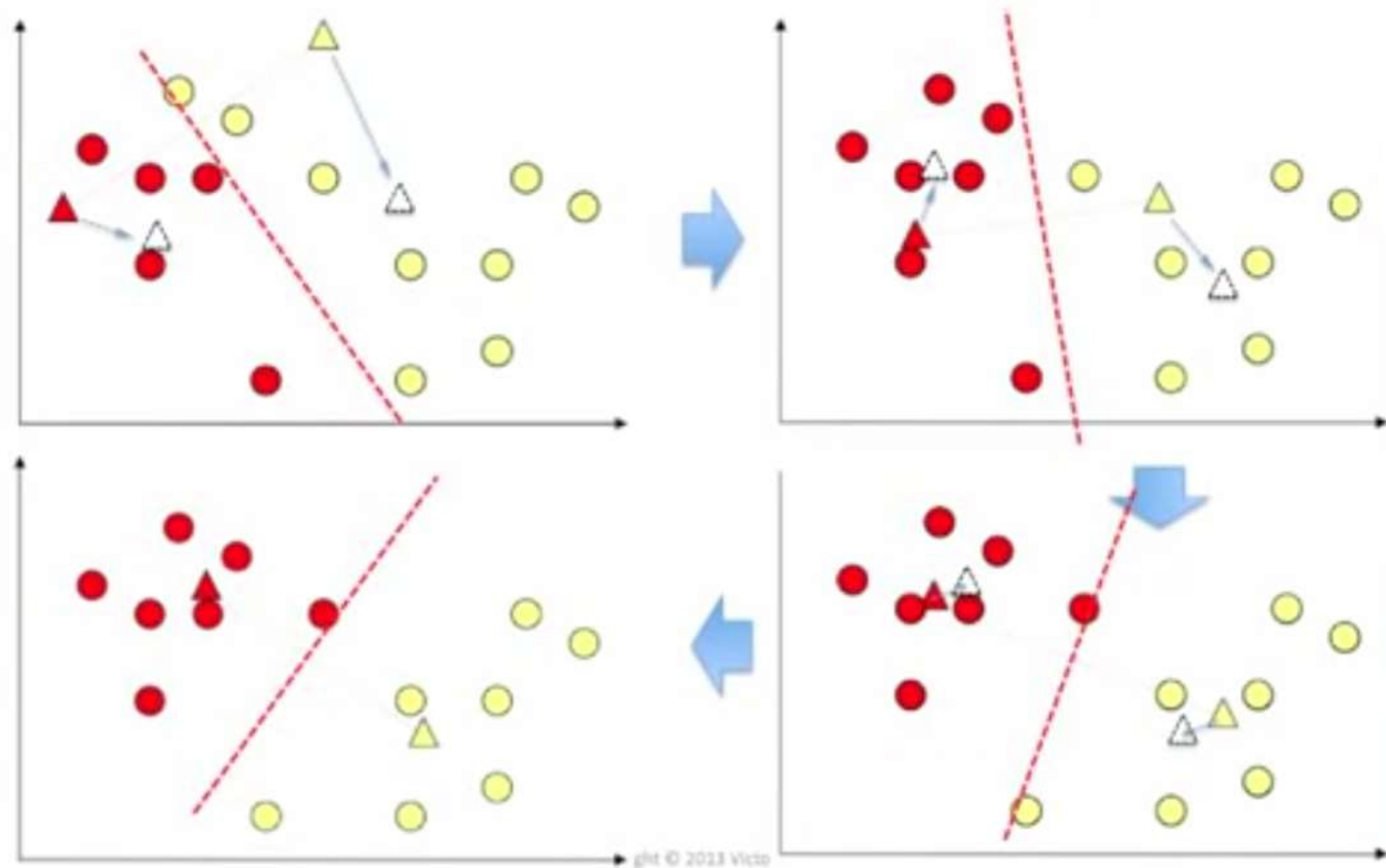
$$d(m1, m2) = 2.93$$

Final Results



Example 2:





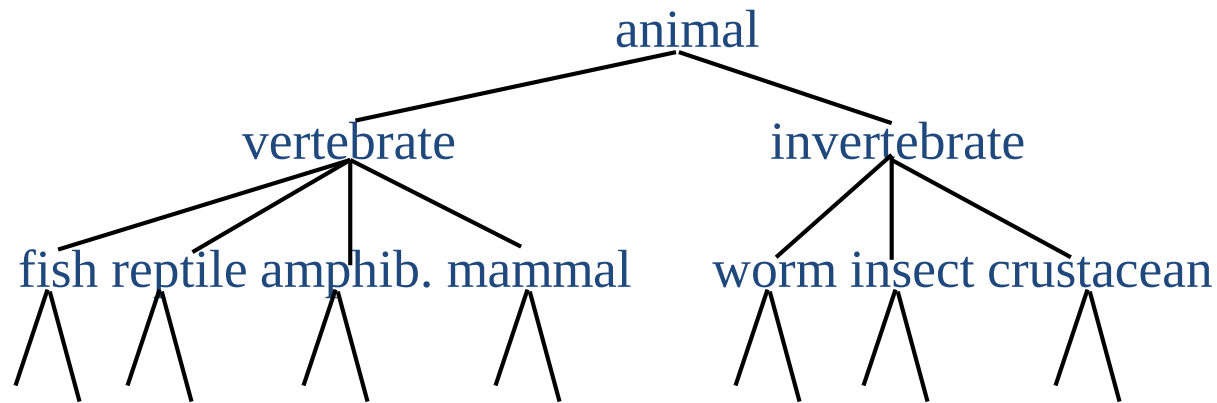
How to decide k?

- Unless the analyst has a prior knowledge of the number of underlying clusters, therefore,
 - Clustering solutions for each value of K is compared
 - The value of K resulting in the smallest SSE being selected

Hierarchical Clustering

Hierarchical Clustering

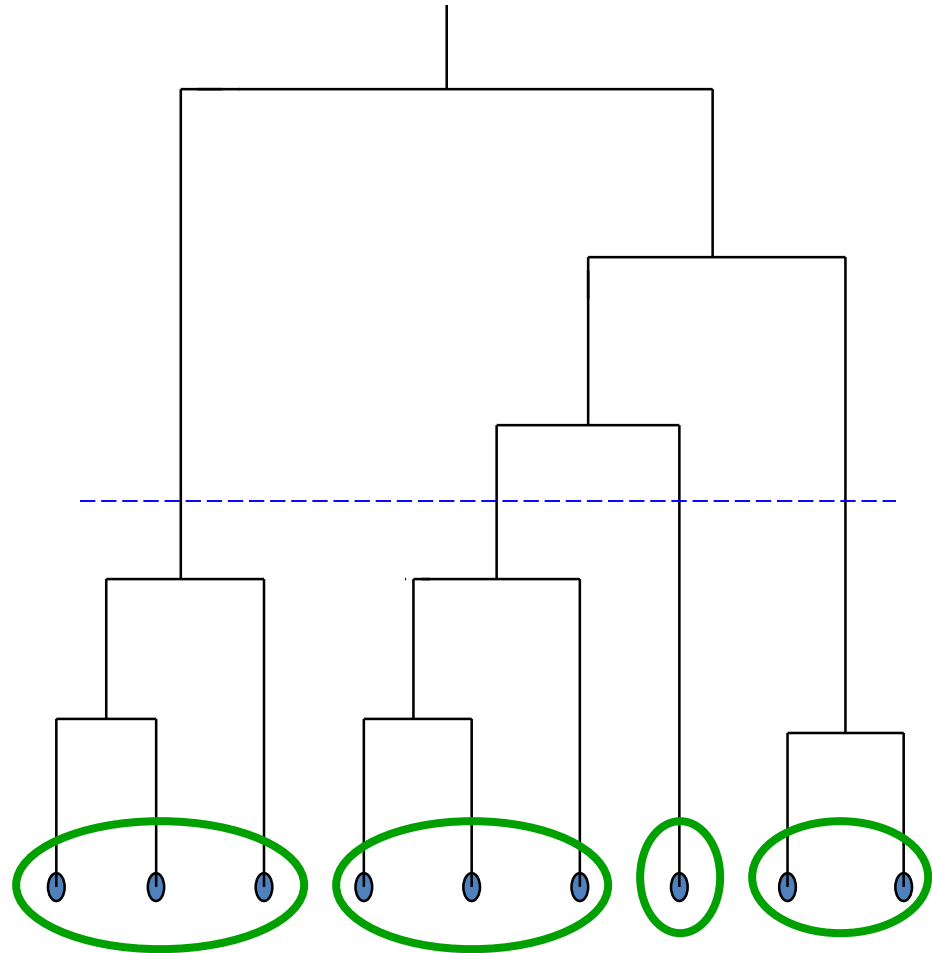
- Build a tree-based hierarchical taxonomy (*dendrogram*)



- *One approach:* recursive application of a partitional clustering algorithm.

Dendrogram: Hierarchical Clustering

- Clustering obtained by cutting the dendrogram at a desired level: each **connected** component forms a cluster.



Hierarchical clustering

- There are two styles of hierarchical clustering algorithms to build a tree from the input set S :
 - **Agglomerative (bottom-up):**
 - Beginning with singletons (sets with 1 element)
 - Merging them until S is achieved as the root.
 - In each steps , the two **closest clusters** are aggregates into a new combined cluster
 - In this way, number of clusters in the data set is reduced at each step
 - Eventually, all records/elements are combined into a single huge cluster
 - It is the most common approach.
 - **Divisive (top-down):**
 - All records are combined in to a one big cluster
 - Then the most **dissimilar** records being split off recursively partitioning S until singleton sets are reached.
- Does not require the number of clusters **k** in advance

Hierarchical Agglomerative Clustering (HAC)

Algorithm

- Assumes a similarity function for determining the similarity of two instances.
- Starts with all instances in a separate cluster and then repeatedly joins the two clusters that are most similar until there is only one cluster.

Start with all instances in their own cluster.

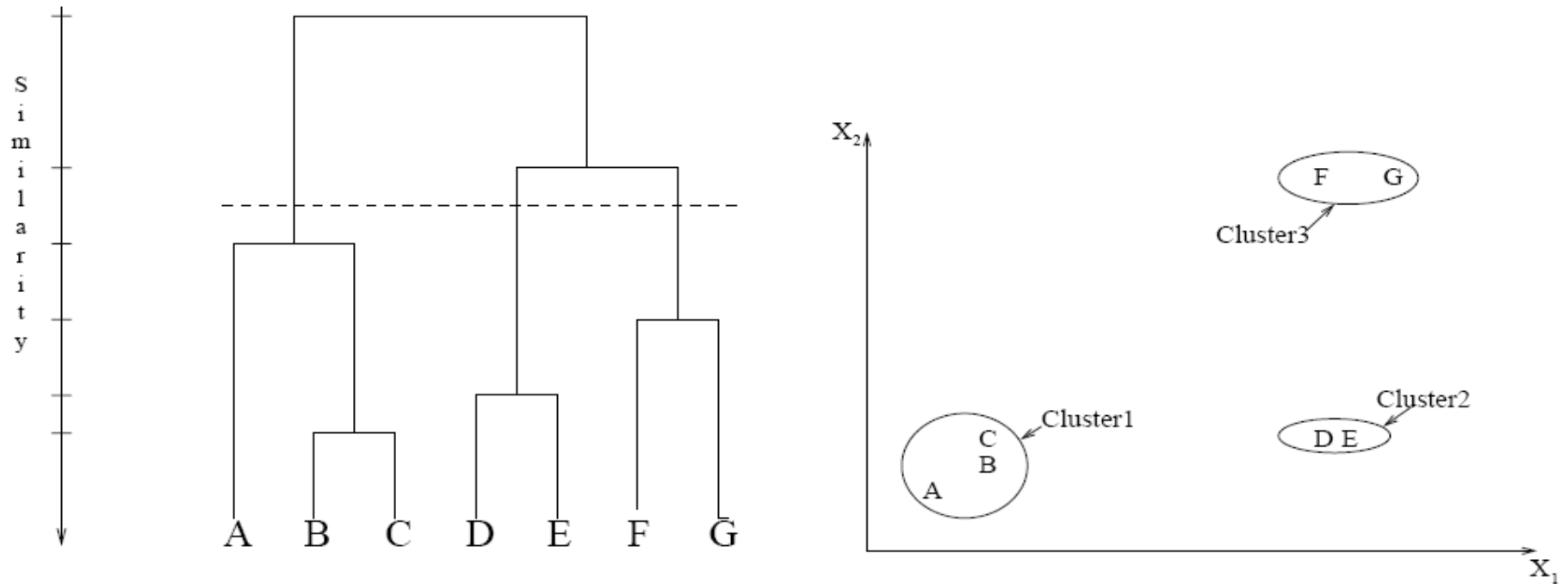
Until there is only one cluster:

Among the current clusters, determine the two clusters, c_i and c_j , that are most similar.

Replace c_i and c_j with a single cluster $c_i \cup c_j$

Hierarchical clustering: forming clusters

- Forming clusters from dendograms



Summary

➤ **K-means**

- **K-means** algorithm is a simple yet popular method for clustering analysis
- *Low complexity :complexity is $O(nkt)$, where $t = \text{\#iterations}$*
- Its performance is determined by initialisation and appropriate distance measure
- There are several **variants** of *K-means* to overcome its weaknesses
 - **K-Medoids**: resistance to noise and/or outliers(data that do not comply with the general behaviour or model of the data)
 - **K-Modes**: extension to categorical data clustering analysis
 - **CLARA**: extension to deal with large data sets
 - Gaussian Mixture models (**EM algorithm**): handling uncertainty of clusters

➤ **Hierarchical Clustering**

- Advantages
 - Dendograms are great for visualization
 - Provides hierarchical relations between clusters
- Disadvantages
 - Not easy to define levels for clusters
 - Experiments showed that other clustering techniques outperform hierarchical clustering